Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 2026
Team ID	
Project Name	FORMULA 1 Race Intelligence Dashboard Analysis Report (2010-2026)
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1.	Programming / Query Language	Language used to query, filter, transform, aggregate, and analyze data.	Python / SQL	Optional preparation and validation; SQL for repeatable aggregations if warehouse used.
2.	Database / Data Warehouse	Warehouse Used to store and manage structured business data for analysis and reporting.	Excel workbook / optional Snowflake	Source workbook contains five analytical sheets; Snowflake is optional, not assumed deployed.
3.	Data Ingestion	Used to load data from source files or external systems into the analytical environment.	Tableau Desktop / Tableau Public	Connect to Excel and relate sheets using race_id, circuit_id, season and driver/team as appropriate.
4.	Data Transformation and Business Logic	Tools/Logic used to clean, join, filter, aggregate, data for analysis and calculate KPIs, metrics, business rules, and analytical measures.	Tableau calculated fields / Data Source page	Type fields, check nulls/duplicates, build KPIs and season/team/driver aggregations.
5.	Data Visualization / Analytics	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	Tableau worksheets and dashboards	Build driver, constructor, tyre, podium, circuit and reliability visualizations.
6.	Front-End / User Interface	Interface through which users interact with dashboards, filters, parameters, and visual insights.	Tableau Desktop / Tableau Public	User-facing interactive dashboards and story points.

Project-specific Documentation

Dataset workbook: F1 Racing Project Dataset(1).xlsx.

Sheets and data rows (excluding headers): race_results 6,464; circuits 29; championship_standings 311; lap_intelligence 3,814; qualifying_results 3,130.

Observed race_results coverage: seasons 2010–2026; 334 distinct race_id values; 19 distinct driver labels; 10 distinct team labels; 21 distinct country labels.

Available analysis fields include race date/season/round, circuit type and location, driver/team, finish/grid position, points, DNF and DNF reason, weather/wet-race flag, safety-car/VSC/red-flag data, pit stops, tyre strategy/starting compound, overtakes and laps led.

Lap-level fields include lap/sector times, tyre compound and tyre age, DRS, speed trap and pit-lap flag. Qualifying fields include qualifying position, Q1/Q2/Q3 times, gap to pole and top speed.

Validation note: workbook is a multi-sheet analytical dataset, not necessarily official complete historical timing. Verify values and data types before presenting findings. Do not infer driver nationality: no nationality field is present.

Use COUNTD([race_id]) for race-event counts where appropriate; race_results contains multiple driver-result rows per event. Distinguish race-result row counts from unique Grand Prix counts.

• Recommended analysis: driver points; constructor season trajectory; tyre compound usage; tyre age versus lap time; podium distribution; country/circuit comparisons; DNF by team/season/reason; wet-race/circuit-type reliability; grid-to-finish position gain; pit-stop and safety-car context.

• Limitations: no driver nationality field; DNF reasons are sparse; dataset's historical coverage and synthetic-looking labels should be validated against an authoritative source before external claims.